



CONNECTED VEHICLE CYBER-RESILIENCE

Trust-aware resilience for connected vehicle platoons



Why cooperative driving needs a trust boundary



Physical V2V test vehicles with different onboard platforms

THE OPERATING TENSION

Cooperation adds useful preview

Neighbor position, speed and acceleration support coordinated platooning beyond local sensing alone.

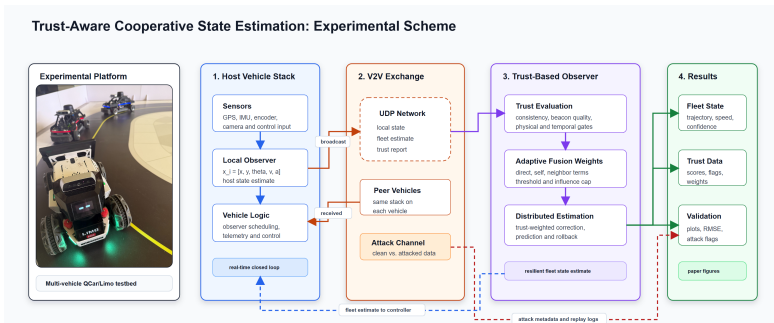
Plausible data can still be harmful

False, replayed, delayed or missing packets can corrupt a fleet estimate before a conventional controller reacts.

Keep cooperation, but reduce dependence on evidence that becomes unreliable.

The trust layer sits between V2V data and the vehicle ECU

PROJECT CONCEPT



Incoming evidence is screened, weighted and corrected before it reaches the existing vehicle logic.

System boundary: the vehicle ECU keeps control

ELECTRONIC BOARD INTEGRATION

Inputs

- IMU and local kinematics
- GNSS position and 1PPS timing
- Vehicle-bus data and V2V packets

Trust edge module

- Timing and packet-validity gates
- State estimation and source weighting
- Rollback, re-anchoring and audit logs

Outputs

- Trusted state estimates
- Attack, trust and health flags
- Safe-mode request through a gateway

Planned path: monitor first; the ACC/CACC ECU retains acceleration and braking authority.

Response to an unreliable packet

ONE PROGRESSIVE RESPONSE CHAIN

01

ACCEPT

Evidence is fresh and consistent.

Prevent new contamination

Packet gate, down-weighting and isolation.

02

DOWN-WEIGHT

Reduce the source's influence.

03

ISOLATE

Set the source weight to zero.

Repair recent history

Rollback and anchoring remove stored error.

04

CORRECT

Roll back and re-anchor state.

Protect vehicle control

Degrade only when confidence is insufficient.

05

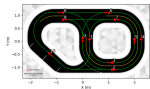
FALLBACK

Request local-control mode.

Current scope: no direct actuator override; the existing vehicle controller keeps command authority.

One loop, three validation environments

VIRTUAL, PHYSICAL AND EMBEDDED VALIDATION



01 Virtual campaign

Numerical and QLab runs repeat the same states and attacks with controlled ground truth.



02 Physical loop

QCar/LIMO trials add real packet timing, sensor noise, scheduling and actuator response.



03 Embedded transfer

The research PCB reproduces the loop with RTOS timing, sensing, communication and logging.

The algorithm and attack scenarios stay comparable while platform realism increases.

The trust logic suppresses corrupted sources in the tested campaign

CORE METHOD - TRUST-AWARE DISTRIBUTED ESTIMATION

Reliability becomes an online fusion weight instead of a fixed assumption.

100%

Detection across the tested attacks and channel configurations

80.64–98.44%

Attacked-source zero rate during the attack window

28–88 ms

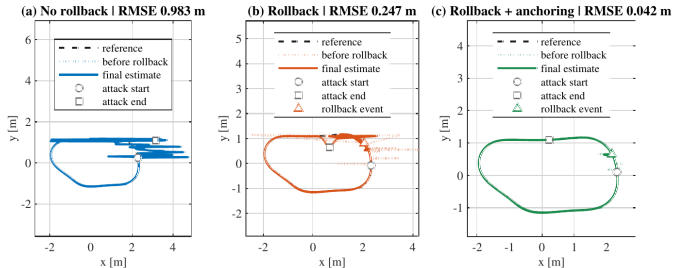
Mean threshold-crossing delay after the 10 s attack onset

Corrupted channel	Mean combined RMSE	Trust drop	Source zero rate
Local	0.643	0.319	80.64%
Global	0.393	0.460	92.90%
Local + global	0.567	0.544	98.44%

Rollback removes contamination already stored in the state

DELAYED-DECISION RECOVERY

XY trajectory comparison for target V0



0.983 m

No rollback

0.247 m

Rollback

0.042 m

Rollback plus accepted
relative-pose anchoring

Local-channel Case 2 RMSE

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Mean attacked-source observer weight

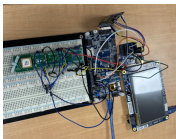
V1 minimum path-projected gap

V2 minimum path-projected gap

Scope: both gaps exceed the 0.30 m

Prototype evolution: breadboard to four-layer PCB

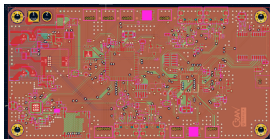
EMBEDDED TRANSFER



01 Bring-up

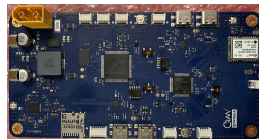
Sensor, display and communication interfaces validated on the bench.

Bring-up evidence: stable 100 Hz RTOS, UART, GNSS acquisition and functional Wi-Fi 6 tests.



02 Layout

Four-layer routing, grounding and power distribution prepared for manufacture.

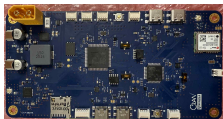


03 Manufactured board

MCU/RTOS platform with sensing, timing, Wi-Fi, storage and development access.

Research PCB vs commercial OBU

DIRECT COMPARISON



Embedded trust demonstrator

Core role: state estimation, source trust, rollback and exported health flags.

Interfaces: IMU, GNSS/1PPS, research Wi-Fi, UART/USB, storage and synchronized logging.

Readiness: proves the embedded trust functions on a research vehicle platform.

Next step: keep the trust layer; add vehicle I/O, production V2X, secure hardware and qualification.



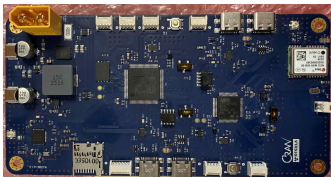
Commercial OBU envelope

Core role: deployment-ready V2X communication, positioning and vehicle integration.

Interfaces: DSRC/C-V2X, cellular/WLAN options, GNSS, CAN/GPIO and onboard sensor inputs.

Readiness: hardware-backed security, automotive design, certifications and lifecycle support.

The next board revision closes four product gaps



Research PCB

Keep the trust, rollback and logging software; redesign the surrounding hardware for the vehicle.

Deployment principle

Begin monitor-only behind a gateway before any fallback request is authorized.

PRODUCTION-ORIENTED REVISION

01 Protected power and enclosure

12/24 V protection, thermal design, automotive components, EMC and environmental validation.

02 Vehicle and V2X interfaces

CAN/CAN-FD or gateway diagnostics and an external or integrated C-V2X/NR-V2X modem.

03 Sensors and timing

Automotive IMU, GNSS/1PPS and selected bus signals; optional vehicle-supplied wheel speed, yaw or pose.

04 Security and lifecycle

Secure boot, signed updates, HSM/secure element, protected logs, keys and update tooling.

Research contribution and product readiness are different questions

POSITIONING

What the research demonstrates

- Trust-adaptive V2V fusion instead of fixed source reliability
- Suppression plus rollback/anchoring for future and stored contamination
- Nonlinear 2D platoon tests with local and global channel attacks
- Trusted estimates exposed to a control-availability decision

Current position: research-validated and hardware-demonstrated, but not yet an automotive-qualified OBU or RSU.

What a product must still prove

- Automotive interfaces, timing and runtime reliability
- ISO/SAE 21434 cybersecurity evidence and secure lifecycle
- ISO 26262 safety concept when control can be influenced
- EMC, environmental, manufacturing and vehicle qualification

How SEGULA can reuse the work

ENGINEERING VALUE ACROSS CUSTOMER PROGRAMS

01

Reusable assets

Trust and rollback software, embedded reference design, repeatable attacks, digital-twin scenarios and synchronized logs.

02

Client work packages

V2X-resilience studies, HW/SW integration, cyber-robustness campaigns and safety, EMC or vehicle-validation support.

03

Value to test

Less repeated setup, a differentiated OEM/Tier-1 offer and follow-on engineering through integration and industrialization.

Commercial gate: confirm customer demand, integration effort, reuse rate and engineering time saved before making an ROI claim.

Decision requested: launch a monitor-only pilot

RECOMMENDED NEXT GATE

Observe first. Measure. Then decide.

The board calculates trust and recovery decisions and records them, but it does not command the actuators.

Go/no-go evidence

False positives, detection delay, packet loss, recovery, compute load and integration effort.

CONDITIONS TO START

01 Vehicle access

A target program provides bus access and a measurable cooperative-driving use case.

02 Production-like interfaces

Connect the target CAN/CAN-FD gateway and a representative V2X modem.

03 Joint ownership

Hardware, software, cybersecurity and safety owners agree on the evidence and stop criteria.

The pilot decides whether further industrialization is justified.